NED UNIVERSITY OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF SOFTWARE ENGINEERIN
Test 1 Solution

1. **Primary Goals of Each Computer Generation:**
    - **First Generation:** (1940s-1950s) Goal was basic computation, focused on hardware like vacuum tubes.
    - **Second Generation:** (1950s-1960s) Introduced transistors, aiming for improved speed and reliability.
    - **Third Generation:** (1960s-1970s) Integrated circuits for smaller size and increased processing power.
    - **Fourth Generation:** (1970s-1980s) Microprocessors, aiming for personal computing and networking.
    - **Fifth Generation:** (1980s-present) Focus on artificial intelligence, parallel processing, and natural language processing.
2. **Memory Hierarchy and System Performance:**
    - **Role:** Memory hierarchy consists of different levels (registers, cache, RAM, storage) with varying speeds and capacities.
    - **Contribution:** Fast, but small, memory (e.g., cache) stores frequently accessed data for quick retrieval, reducing CPU wait times.
    - **Impact:** Efficient use of the hierarchy optimizes data access, improving overall system performance.

      By using memory hierarchy we can improve the performance and fulfill the storage requirement in a reasonable cost.
3. **Computers in Healthcare:**
    - **Utilization:** Computers aid in patient records (EHRs), medical imaging (MRI, CT scans), drug discovery, and treatment planning.
    - **Examples:** Picture Archiving and Communication System (PACS) for medical imaging, Clinical Decision Support Systems (CDSS), telemedicine.
4. **First Generation vs. Subsequent Generations:**
    - **Technology (First Generation):** Used vacuum tubes, large size, high power consumption.
    - **Technology (Subsequent Generations):** Introduced transistors, integrated circuits, and microprocessors, leading to smaller size, increased speed, and reduced power consumption.
    - **Capabilities (First Generation):** Basic calculations, limited programming languages.
    - **Capabilities (Subsequent Generations):** Advanced programming languages, improved computational power, increased versatility.
5. **Cache Hit Ratio Calculation:**
    - **Given:** Cache misses = 2,000, Total memory accesses = 5,000.
    - **Formula:** Cache Hit Ratio = (Total accesses - Cache misses) / Total accesses.
    - **Calculation:** (5,000 - 2,000) / 5,000 = 3,000 / 5,000 = 0.6 or 60%.

NED UNIVERSITY OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF SOFTWARE ENGINEERIN
Test 1 Solution

1. **Locality's Influence on Memory Hierarchy:**
    - **Temporal Locality:** Refers to the reuse of recently accessed data. It influences cache design, emphasizing storing recently used data for quick retrieval.
    - **Spatial Locality:** Focuses on the tendency of accessing neighboring memory locations. It influences the design of cache lines, ensuring that blocks of contiguous data are loaded together for efficiency.
2. **Evolution of Computers Across Generations:**
    - **First Generation:** (1940s-1950s) Vacuum tubes for basic calculations.
    - **Second Generation:** (1950s-1960s) Transistors brought speed and reliability.
    - **Third Generation:** (1960s-1970s) Integrated circuits allowed smaller size and increased processing power.
    - **Fourth Generation:** (1970s-1980s) Microprocessors enabled personal computing and networking.
    - **Fifth Generation:** (1980s-present) Focus on artificial intelligence, parallel processing, and natural language processing.
3. **Computers in Education:**
    - **Integration:** Computers are used for online learning, educational software, and collaborative tools.
    - **Examples:** Learning Management Systems (LMS), Interactive Whiteboards, Educational Games, and Virtual Reality (VR) for immersive learning.
4. **Importance of Multiple Levels of Memory:**
    - **Fast Access:** Levels like cache provide fast access to frequently used data, reducing CPU wait times.
    - **Capacity:** Higher levels like RAM offer more capacity, accommodating larger amounts of data.
    - **Hierarchy:** Allows optimization for both speed and capacity, balancing the trade-off between fast but limited and slower but larger memory.
5. **Cache Hit Ratio Calculation:**
    - **Given:** Cache misses = 4,000, Total memory accesses = 6,000.
    - **Formula:** Cache Hit Ratio = (Total accesses - Cache misses) / Total accesses.
    - **Calculation:** (6,000 - 4,000) / 6,000 = 2,000 / 6,000 = 1/3 or approximately 0.3333, representing a cache hit ratio of 33.33%.